

Application Programming Interface (API) description

REST webservice "Ordersimulate"

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Created by

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1. Contents of this documentation

This documentation describes the REYHER "Ordersimulate" interface and the required "Request Authentication Token" interface. Using "Order Simulate", customer-specific prices and availability can be queried automatically.

2. Basic information

The functionality of these interfaces is provided as a RESTful web service. Access to the service is exclusively available via encrypted HTTPS connections. Message content (payload) must be transmitted in JSON format via HTTP POST.

2.1. Limitation of requests and limits

The aim of this interface is not to query the entire REYHER range, which is why the number of daily requests is limited to 300 and the number of items per request to 100. A minimum time interval of 5 seconds must be maintained between two requests. If these limits are not adhered to, a corresponding error message is returned (see chapter "[Status codes](#)").

REYHER reserves the right to further restrict the service in the event of improper use.

2.2. Validation of the message content

The transmitted message content (payload) is validated on the server side to check the syntax used. If validation fails, a corresponding error message is returned (see chapter "[Status codes](#)").

2.3. Token-based authentication

The authentication of the request takes place via a (bearer) token sent in the header of the request. The token can be requested via an additional interface (see section "[Interface description Request Authentication Token](#)") by transferring access data previously provided by REYHER. Alternatively, the token can also be initially provided to the user by REYHER. A requested token is currently valid for 10 years or until revoked.

REYHER reserves the right to limit the validity period of the token during the course of the project. In this case, a new token must be requested regularly via the token interface.

Authentication at the "Ordersimulate" interface is carried out by transferring the bearer token in the "Authorization" header field of the request:

```
Authorization: Bearer <authentication token>
```

3. Interface description "Request Authentication Token"

3.1. General information

This interface is used to request a token. This token is required for authentication at the Ordersimulate interface. Access data must be provided in order to create the token. You will receive the access data from your REYHER contact person. Tokens are currently valid for 10 years or until revoked.

URL of the interface:

```
https://rio.reyher.de/rest/reyher_int_de/V1/integration/customer/token
```

3.2. Fields of the JSON message (request)

Field	Description	Example
username	ID of the user in RIO	example.test@domain.com
password	Password of the user	password123

3.3. Example request

```
{  
  "username": "example.test@domain.com",  
  "password": "password123"  
}
```

3.4. Response example

```
"eyJraWQiOiIiIiwiaWV4bnIjoisSFMyNTYifQ.eyJ1aWQiOiI3OTA2MiwidXR5cGlkIjozLCJpYXQiOiIyMDI0MjE2MjgzMzEsImV4cCI6MjAxMjk4ODMzMX0.Kawp_aQVa-cahyLzYVucpc0ubkAfv2x_LKAn9zKCkd8"
```

4. Interface description "Ordersimulate"

4.1. General information

This interface is used for automated price and availability queries. Based on the REYHER-SKU, customer-specific prices and availabilities can be queried for individual or multiple items.

URL of the interface:

```
https://rio.reyher.de/rest/reyher_int_de/V1/sapordersimulate/mine/getOrdersimulate
```

4.2. Fields of the JSON message (request)

Field	Description	Example
items	Parent element	
position	A unique item number must be transferred for each requested item in a request. This field is used to identify the returned items.	1
sku	In this field, the REYHER article number (15 digits) must be transferred as the identifier of the article to be checked.	015873000030000
quantity	Requested quantity. If a quantity not equal to one PU is specified, the quantity will be rounded up to the next full PU quantity for processing the request. If no value is transferred in this field or the field is missing in the request, one PU is assumed as the basis for processing the request.	1000

4.3. Example request

```
{
  "items": [
    {
      "position": 0,
      "sku": "015873000030000",
      "quantity": 1000
    },
    {
      "position": 1,
      "sku": "015873000140000",
      "quantity": 1000
    }
  ]
}
```

4.4. Fields of the JSON message (response)

Field	Description	Example
OrderSimulateResponse	Parent element	
created	Time of processing	2023-10-18T12:05:43.000000+00:00
Statusinformation	see chapter " Status codes "	0
Payload	Parent element	
Items	Parent element	
Position	Identifier from the request	1
SKU	SKU (may differ from the requested SKU)	015873000030000
Price	Price per 100; excluding global discounts and surcharges	27.48
PriceQuantity	Quantity basis for the "Price" field	100
PositionPrice	Price for the requested quantity; including global discounts and surcharges	287.16
PositionPriceQuantity	Quantity basis for the "PositionPrice" field	1000
QuantityUnit	Unit of measure	ST
QuantityAvailable	Available quantity based on the requested quantity	1000
Currency	Currency	EUR
Remark	Unused field	

4.5. Example response

```
{
  "OrderSimulateResponse": {
    "created": "2023-10-19T08:28:31.000000+00:00",
    "Statusinformation": "OK",
    "StatusCode": 0,
    "Payload": {
      "Items": [
        {
          "Position": 0,
          "SKU": "000015873000030000",
          "Price": 14.1,
          "PriceQuantity": "100",
          "PositionPrice": 147.3,
          "PositionPriceQuantity": "1000",
          "QuantityUnit": "ST",
          "QuantityAvailable": 1000,
          "Currency": "EUR",
          "Remark": ""
        },
        {
          "Position": 1,
          "SKU": "000015873000140000",
          "Price": 160.86,
          "PriceQuantity": "100",
          "PositionPrice": 1680.93,
          "PositionPriceQuantity": "1000",
          "QuantityUnit": "ST",
          "QuantityAvailable": 900,
          "Currency": "EUR",
          "Remark": ""
        }
      ]
    }
  }
}
```

5. Status codes

StatusCode	Status information	Description
0	OK	Request was processed within normal parameters
10	Unknown sku	One of the SKUs cannot be assigned to a product
20	An internal error occurred	An internal error has occurred during the request
21	an internal error occured during order simulation	An error has occurred during the ERP process
22	Invalid syntax	The requested JSON is invalid
30	Too many requests today	The number of permitted requests for this user has been exceeded
31	Waiting period since last request too low	Too little time has passed since the last request
32	Too many items requested	More than 100 articles were requested
40	Authentication failed	Authentication has failed
50	API disabled	The interface is currently not available

6. Contact person

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